PROJECT-BASED LEARNING

PHASE-II EVALUATION

CYBER CRIME CASE MANAGEMENT SYSTEM

Team ID: DSCPP-III-2026-T093

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27

Team & Mentor Details

Team Information

Team ID: DSCPP-III-2026-T093
Semester: 3rd
Section: CORE-F/ML-1/2/4
Domain: AI/ML

Team Members

1. Kusum Sapra – 2510011573
2. Ananya Singhal – 2510370189
3. Rudransh Garg – 2510370238
4. Aryan Gusain – 2510370533

Mentor

Ms.Ayushi Saini **Interactions completed: 6**

Project Overview

Our team is developing a Cyber Crime Case Management System to organize cybercrime case records and investigation information. The work covers data structures, C++ OOP, MySQL database management, and system integration.

Phase-I Recap & Feedback Incorporated

Phase-I Status

- Identified the need for organized cybercrime case management.
- Defined cases, users, officers, evidence, investigations and reports.
- Selected C/C++ for data structures and OOP concepts.
- Selected MySQL for persistent storage.
- Established a modular development plan.

Feedback Received

- Improve data organization
- Improve system design
- Ensure persistent storage

Changes Incorporated

- Added data structures for case processing
- Divided project into C, C++ and database modules
- Implemented MySQL tables and database operations

Phase-II Progress

Phase I established the problem and initial design. During Phase II, the project progressed toward implementation through data structures, OOP classes, database design and integration testing.

Problem Statement & Finalized Objectives

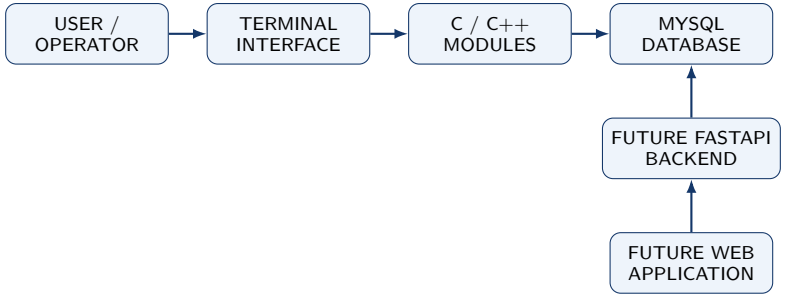
Problem Statement

Managing cybercrime cases involves maintaining case details, assigning investigating officers, tracking evidence, recording investigation activities and preparing reports. Without a structured system, searching, updating and organizing these records becomes difficult.
The project aims to develop a modular Cyber Crime Case Management System using C, C++ and MySQL, with a backend API and web interface planned for subsequent development.

Finalized Objectives

1. Implement data structures for efficient case organization and processing.
2. Apply C++ OOP concepts to model cases, users, officers, evidence and investigations.
3. Store and manage records using a relational MySQL database.
4. Develop an integrated system that can later support a backend API and professional web interface.

Updated System Architecture / Workflow



Major Components

- Input: Terminal case and record input
- Processing: C data structures and C++ classes
- Database: MySQL tables and relationships
- Output: Case details, searches and related records

Changes Since Phase-I

- Implemented core data structures and OOP classes
- Added MySQL persistence
- Tested C/C++ and MySQL integration
- Planned FastAPI and web frontend



Module Design & Functional Requirements

Major Modules

1. **Case Management**
   - Create, display, search, update and delete cases
   - Maintain ID, victim, crime type, priority and status
2. **Data Structures**
   - Linked list, queue, priority queue, stack and hash table
3. **Officer & Investigation**
   - User/Officer classes, assignment, evidence and investigations
4. **Database Management**
   - MySQL storage and CRUD operations

Functional Requirements

- Add, search, update and delete records
- Assign officers to cases
- Store and retrieve evidence
- Store investigation history
- Create and retrieve reports
- Preserve records in the database

Module Dependency

Case management uses the data structures and case model. Officer assignment connects cases with officers. Evidence, investigation history and reports reference cases through database relationships.

Technology Stack & Development Methodology

Technology Stack

- Programming: C and C++
- Database: MySQL 8.0
- Connectivity: MySQL Connector/C++ 26.7
- Compiler: Microsoft Visual C++ (MSVC)
- Tools: VS Code, Terminal, MySQL Workbench
- Version Control: Git and GitHub
- Planned Backend: Python with FastAPI
- Planned Frontend: React with Vite

Development Methodology

1. Requirement analysis
2. Architecture and database design
3. Individual module development
4. Module-level testing
5. C/C++ and MySQL integration
6. Backend/API development
7. Frontend development and final validation

Implementation of PBL Course Concepts

Course	Concepts Implemented	Project Module
Data Structures in C	Linked list, queue, priority queue, stack and hashing	Case processing and lookup
OOP with C++	Classes, objects, constructors, encapsulation, inheritance, virtual functions and polymorphism	Case, User, Officer, Evidence, Investigation
DBMS	SQL tables, primary keys, foreign keys, CRUD and relational queries	MySQL database module
Operating Systems	No specific OS concept currently established as a project feature	Not claimed as implemented

Technical Implementation

- **Linked List:** Dynamically allocated case nodes.
- **Queue:** FIFO processing.
- **Priority Queue:** Priority-based case processing.
- **Stack:** LIFO activity processing.
- **Hash Table:** Hashing with linear probing for case ID lookup.
- **OOP:** Officer inherits from User; virtual functions support polymorphism.
- **DBMS:** Related records stored across project tables.

Only concepts actually implemented and demonstrable should be claimed.

Development & Implementation Progress

Module	Status	Work Completed
C Data Structures	Completed	Linked list, queue, priority queue, stack and hash table
C++ OOP	Completed	Case, User, Officer, Evidence and Investigation classes
MySQL Database	Completed	Schema, tables, CRUD operations and database connection
Database Integration Tests	Completed	Case CRUD, officer management, assignment, evidence, investigations and reports
Terminal Application	In Progress	Core modules exist; consolidated workflow remains
FastAPI Backend	Pending	API endpoints and database integration planned
Web Frontend	Pending	Planned for the final phase

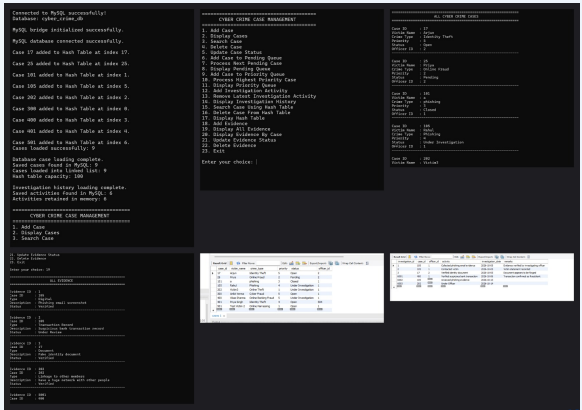
Overall Project Completion: *70%*

Working Prototype / Demonstration

Implemented Features

- Dynamic case records using linked list
- Case search, update and deletion
- Queue, priority queue, stack and hash table
- C++ Case and User classes
- Officer inheritance from User
- MySQL case CRUD
- Officer creation, search and availability update
- Officer case assignment
- Evidence and investigation operations
- Report database operations

Prototype / Output Evidence



Testing, Results & Validation

Test / Module	Expected Result	Observed Result	Status
Linked List	Add, display, search and delete nodes	Module implemented	Completed
Hash Table	Insert and search case IDs	Linear probing implemented	Completed
C++ OOP	Display case and officer details	OOP classes implemented	Completed
MySQL Connection	Establish database session	Connection test succeeded	Pass
Case CRUD	Insert, retrieve and update cases	Integration test succeeded	Pass
Officer Management	Add/search officer and update availability	Officer test succeeded	Pass
Officer Assignment	Link officer with case	Assignment test succeeded	Pass
Evidence / Investigation	Store and retrieve related records	Integration test succeeded	Pass
Reports	Create and retrieve reports	Integration test succeeded	Pass
FastAPI	Return API responses	Not yet implemented	Pending
Web Frontend	Browser-based interaction	Not yet implemented	Pending

Results / Validation

The demonstrated MySQL connection, case CRUD, officer management, assignment and database integration tests have passed. API performance and frontend testing will be reported only after actual implementation and measurement.

GitHub Repository & Development Activity

Repository Information

- **Repository:** `https://github.com/kusumsapra/CYBER_CRIMECASE_MANAGEMENT`
- Branch: master
- Version control: Git and GitHub
- Main modules: C_Module, Queue, PriorityQueue, Stack, HashTable, CPP_Module, Database, Data, main.c
- Planned module: backend/

Development Activity

- Source code maintained in repository
- Data structures, OOP and MySQL integration
- Database testing and validation
- Meaningful commits for features and fixes
- .gitignore for generated files
- All four members as contributors
- Individual commits using own GitHub accounts

Individual Contribution

Member	Role	Technical Contribution
Ananya Singhal	OOP Development	Case, User, Officer, Evidence and Investigation classes
Rudransh Garg	Database Development	MySQL schema, table relationships, CRUD and database management
Kusum Sapra	Data Structures	Linked List,queue,priority queue,stack and hash table
Aryan Gusain	Integration Testing	C/C++ integration, database test programs and validation

Challenges, Remaining Work & Phase-III Roadmap

Challenges Faced

- MySQL Connector/C++ and compiler compatibility
- Header, library and runtime DLL configuration
- C/C++ integration without object-file conflicts
- Maintaining consistency between application and database records

Solutions / Learning

- Configured MSVC Build Tools
- Compiled C/C++ modules separately
- Tested database features through separate programs
- Used relational keys between project entities

Phase-III Roadmap

1. Complete consolidated terminal application
2. Develop FastAPI backend with MySQL
3. Implement and test API endpoints
4. Add authentication, validation and error handling
5. Develop professional web frontend
6. Integrate frontend and backend
7. Final testing and documentation

Expected Outcomes & References

Expected Final Outcomes

- Working Cyber Crime Case Management System
- Data structures for case processing
- C++ OOP models for system entities
- MySQL persistence and relational data management
- Terminal-based demonstration interface
- FastAPI backend and REST API
- Professional web frontend
- Testing, documentation and integrated application

References

1. C programming language documentation
   `https://en.cppreference.com/w/c`
2. C++ language reference
   `https://en.cppreference.com/w/cpp`
3. MySQL 8.0 Reference Manual
   `https://dev.mysql.com/doc/refman/8.0/en/`
4. MySQL Connector/C++ documentation
   `https://dev.mysql.com/doc/connector-cpp/en/`
5. FastAPI documentation
   `https://fastapi.tiangolo.com/`
6. Git documentation
   `https://git-scm.com/doc`

Thank You

Questions & Suggestions